A Critique of Surprise in Generative Art

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Abstract

In contemporary generative art, the aesthetics of surprise has become a central element in both the production and reception of artwork. While sciences that investigate chaotic or emergent phenomena attempt to generate the unexpected in order to explain and manage it, generative art moves in the opposite direction, cultivating surprise as a foundation for aesthetic experience. Understanding that the experience of surprise is influenced by larger cultural and economic factors, however, is essential for a generative artist's self-reflective, critical practice. This paper foregrounds the importance of surprise in generative art and argues that it mirrors neoliberal capitalism, training us to seek surprise and to find pleasure in its repeated production. This argument runs counter to recent theoretical work that attempts to position generative art as a methodology beyond history and the influence of ideology.

Heading: Generative Art and Surprise

At the end of his influential essay "What is Generative Art?" Philip Galanter claims that "Generative art is ideologically neutral. It is simply a way of creating art and any content considerations are up to the given artist." [1] Such a claim undermines critical perspectives that take into account particular historical periods and neutralizes ideological analysis itself. The post-historical generative artist can simply "make art" without worrying that her choices are motivated by unconscious ideologies or influenced by her historical period. In such a worldview, the search for surprise is simply one artistic goal in a plurality of possibilities—not a widespread, constituent aspect of the practice that can be critically analyzed in terms of ideological motivations. Accepting this view, nothing would remain to unite the widespread recourse to surprise in generative art practices today, except perhaps the attempts by artists to emulate unpredictable scientific or natural processes, or as Galanter calls them, "naturally occurring processes beyond the influence of culture and man." [2] The popularity of models of natural processes—flocking, schooling, fluids, colonies—in contemporary generative art

suggests the widespread embrace of such emulation where surprise simply becomes a 'natural' effect through which the unexpected may arise. In this essay, we explore elements of contemporary surprise, drawing links between surprise in neoliberal capitalism and in generative art, arguing that surprise, like all aesthetic criteria, is deeply influenced by our historical situation and the ideologies it embeds.

What is surprise, and how does it differ from other human sensory and emotional experiences? In 1795, Adam Smith, a key figure of capitalist economics, wrote "[w]hat is new and singular, excites that sentiment which, in strict propriety, is called Wonder; what is unexpected, Surprise; and what is great or beautiful, Admiration." [3] For Smith, surprise is the unexpected, or the "violent and sudden change produced upon the mind." [4] In fact, we often speak of 'being taken' by surprise, a phrase that emphasizes a loss of control when confronted with the surprising. Surprise ('sur-prehend') literally 'over-takes' us, transporting our experience in a way that seems beyond our control. Yet, today, surprise is not completely unexpected, and we are trained to expect the unexpected, to anticipate the new, and even to be astonished at its arrival. The dynamics of capitalism foreground novelty and innovation to the extent that that we now expect the continual arrival of new products, new technologies, and new experiences; like a Jack-in-the-Box toy, surprise is hardly surprising at a fundamental level. Instead, the conditions and limits within which surprise can function are carefully delimited and controlled. Within this strict frame of sanctioned surprise, we are trained to find joy and wonder, but it is not a surprise that is ever threatening to the conditions by which it was created.

Notions of surprise, the unexpected, and the unpredictable often appear in writings on generative art, in some cases even appearing as constitutive of the practice. In his book, *Generative Art*, Matt Pearson offers two rules which define the core principles of the generative method, one of which invokes the importance of surprise:

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> To be able to call a methodology generative, our first hard-and-fast rule needs to be that *autonomy must be involved*. The artist creates the ground rules and formulae, usually including random or semi-random elements, and then kicks off an autonomous process to create the artwork. The system can't entirely be under the control of the artist, or the only generative element is the artist herself. The second hard-and-fast rule therefore is *there must be a degree of unpredictability*. It must be possible for the artist to be as surprised by the outcome as anyone else. [5]
> **[BLOCK QUOTE ENDS]**

The element of surprise here is not unrelated to the first aspect of the definition, for the loss of control is intimately connected to the possibility of 'being taken' by surprise. As the artist Raphael Lozano-Hammer explains, "most electronic artists are looking for an out-of-control

quality that will result in their work actually having outcomes that they did not anticipate. If the piece does not surprise the author in some way then it is not truly successful in my opinion." [6] The decision to cede aesthetic control to a mechanism or system can be understood as motivated by the structure of surprise itself: if one wants to be overcome by something, then the first step is to create this "something" which can overcome. Thus, the loss of control—constitutive of surprise—is often a core element of generative methods; a feature that Inke Arns has called the "*negation of intentionality*." [7]

This negation is foregrounded in Galanter's own definition of generative art:

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> Generative art refers to any art practice where the artist uses a system, such as a set of natural language rules, a computer program, a machine, or other procedural invention, which is set into motion with some degree of autonomy contributing to or resulting in a completed work of art. [8]
> **[BLOCK QUOTE ENDS]**

Though surprise is not mentioned, it is still essential for Galanter, who correlates it with information theory, stating that "the more 'surprise' a given communication can exhibit the more information it contains." [9] To demonstrate, he uses the example of a program that outputs a string of identical letters. When the next letter appears, we can easily predict the outcome and are not surprised; the information content of the message is low and appears highly ordered (low entropy). At the other end of the spectrum is a program that outputs completely random letters. Since we can never predict what letter will appear, the information we receive from the sequence is high but also disordered (high entropy). Crucially for information science, and thus for Galanter, the completely random string of letters—though unpredictable—does *not* surprise us, as we have come to expect this unpredictability. This idea can be traced back to Wolfram's classes of Cellular Automata from 1984 and to Langton's 1986 experiments with his 'λ (lambda) parameter'. [10] [11] Building on Wolfram's work, Langton located what he called the 'balanced' region of emergence between 'quiescent' (stable) and 'chaotic' (unstable) regions and argued that this region offered "the most promise for supporting an artificial bio-chemistry." [12] In such a region we find not the absence of surprise we see in pure repetition, nor the unsurprising surprise of pure randomness, nor the inadvertent surprise of human error, [13] but what artist Marius Watz has referred to as "genuine surprise." [14]

Importing these ideas into an art context, Galanter writes that "in terms of our human ability to extract meaning from a given experience we require a mix of surprise and redundancy, i.e. a signal somewhere between extreme order and disorder." [15] In what we might call an informatic surprise, too much order *or disorder* is experientially predictable and erases the meaning that can occur in the region between these poles. In such accounts, informatic surprise fades at the opposite poles of a measurable continuum from repetition to

randomness, and a middle ground is privileged for so-called "complex" works combining surprise and redundancy. Information theory is mobilized as a means of quantifying subjective human experience, mapping the outputs of generative systems (and the degree of meaning they can impart) to the levels of order and disorder they exhibit. This impulse—to attempt to locate a quantitative basis for aesthetic judgements—is at least as old as Enlightenment philosophy and its search for universal means with which to judge the success of art. [16] Yet it is the current ideology of neoliberal datafication and its "widespread belief in the objective quantification and potential tracking of all kinds of human behavior," that we perhaps find the clearest expression of this impulse. [17]

Heading: The History of Surprise

From where does the centrality of surprise in generative art—for both creator and audience—originate? This is a critical question that moves beyond ontological forays into discussions of what generative art 'is' and toward its potential external influences. Yet central currents in theorizing generative art have tended to displace critical investigation with an interest in methodology. In the words of McCormack et. al., "in contrast to the critical and social analysis that has traditionally surrounded art movements, generative art is understood primarily as a methodology, with little, if anything, said about the art itself or the motivations of its practitioners." [18] This is not to say that theorists and practitioners are uninterested in the history of generative art. One frequently encounters the idea that generative art has an expansive history predating computational experiments. Galanter, for example, argues that "generative art historically precedes modernism, post-modernism, and just about any other 'ism' on record," while Pearson argues that "generative art may not be quite as old as art itself, but it may be said to be at least as old as mathematics." [19] Focusing on the generative method as a quasi-timeless practice for the creation of art again attempts to provide justification for a version of generative art that transcends historical, cultural, and social contexts. Yet it is unlikely that theorists of generative art would disagree with the assertion that all art practices are shaped by the social, cultural, and political values of the period. [20] One would imagine that generative art from the past—Islamic tiles for example—would have been be motivated by different reasons than say, early computer-generated art from Manfred Mohr or Vera Molnar. Likewise, one would expect it to be appreciated by an audience for different reasons.

Observations concerning the historical expansiveness of generative art are influenced and motivated by our own historical situation—alive with post-critical, post-ideological, and post-historical trends that still vibrate throughout postmodern culture and politics. These observations reflect an attitude that is popular amongst theorists of digital media forms who have turned from the jingoism of digital media's absolute newness and rupture—from the analog, from materiality—in order to locate a more nuanced and sober assessment of digital media's debt to the past. [21] Of course, such a move can itself be linked to various ideological goals; as an attempt, for example, to legitimate an art practice through recourse to

historical precedent. The suggestion that generative art existed seventy-thousand years ago operates in this fashion, positioning the practice prior to the origins of representational art, as often invoked through reference to the famed Lascaux Cave paintings. [22] The move to history can also be used to de-legitimate or chastise an art practice that has become swollen with its own importance. Thus, intoxicated claims that computers have ushered in a new, technologically vibrant form of art-making are sobered by the realization that analog generative practices have existed for centuries. [23]

Similarly, the meaning of 'surprise', like all artistic criteria, differs according to shifting historical circumstances. [24] For example, Longinus characterized the aesthetic impact of great writing and rhetoric as "[appearing] suddenly; like a thunderbolt it carries all before it and reveals the writer's full power in a flash." [25] This surprising suddenness did not stem from the arrival of the new (which Longinus scorned and which is associated more with the modern era), but with the creation and strengthening of an audience's memory in an oral culture where the thunder of a great expression would reverberate within an individual's mind to create a lasting impression. Or, in the 19th and 20th centuries, techniques of aesthetic shock (e.g. Charles Baudelaire, Dada) and later, estrangement (e.g. Viktor Shklovsky, Bertolt Brecht) were modes of surprise that responded to the increased rationalization of industrial society, and, as Philip Fisher writes, had "everything to do with the problem of boredom, with overfamiliarity, with the dull effects of habit." [26] The point is that the nature and meaning of surprise shift historically. Longinus' version of the effects of surprise differs from 19th and 20th century conceptions that attempt to undermine bourgeois sensibility or to question capitalism's focus on new desires, experiences, and commodities.

What about the aesthetics of surprise today? We have already noted how generative artists and theorists embrace notions of complexity, creating works in which informatic surprise is foregrounded. Yet it is not immediately clear why surprise functions so often today as an artistic value in itself. Jon McCormack and Alan Dorin's concept of the "computational sublime" is helpful here. [27] They argue that in generative computational work, "even though we cannot comprehend the process directly, we can experience it through the machine—hence we are forced to relinquish control. It is possible to realize processes of this kind in the computer due to the speed and scale of its internal mechanism, and because its operations occur at a rate and in a space vastly different to the realm of our direct perceptual experience." [28] The computational sublime constantly prompts us to be astonished by the supra-sensible powers of the machine. We do not have access to the imperceptible workings and blistering speed that give rise to their surprising effects. Perhaps when confronted with the unexpected performance of a machine, our surprise is transformed into what Smith called "admiration", which augments our embrace of computer technology. In the 18th and 19th century sublime, the experience of the sublime concerned the human. When the human imagination ran up against its limits and humans lost control, reason—via science and mathematics—stepped in to reassert it. If, for example, contemplating the innumerable grains of sand on a beach brought our imagination to its limits, our faculty of reason was

able to assert the idea of infinity. The surprise found in such experiences of the sublime suggests that the human transcends brute nature; that we are more than simply sensuous beings. We have mind, morals, and reason. In the computational sublime, however, the surprising moment stems not from the realization that humans transcend nature and its mechanisms, but that machines may be transcending the human. "[C]an a machine generate something new, meaningful, surprising, and of value…?" ask McCormack et al. in their article, "Ten Questions Concerning Generative Computer Art." [29] Today, intentionally or otherwise, the search for surprise within generative art seems to answer this question in the affirmative, renewing surprise through our confrontation with powerful computational machines.

Heading: "Surprise, Surprise…"

There may be other influences beyond generating admiration for computers, the emulation of complexity science, or 'the science of surprise' [30], that contribute to the centrality of surprise in contemporary generative art. Generative art and the computational sublime also mirror capitalism's religion of innovation and reinforce our impulse to be surprised by the appearance of the new.

Consider the contradictory idea that to be on the lookout for surprise, to constantly expect the unexpected, would diminish the possibility of surprise actually appearing. Ludwig Wittgenstein's well-known joke is appropriate here: "When I came home I expected a surprise and there was no surprise for me, so, of course, I was surprised." [31] When expecting surprise and actively looking for the unexpected, actual surprise can only occur when failing to encounter surprise altogether. Interestingly, this does not contradict the science of surprise, as scientists seek surprise primarily in order to explain it away, rather than embracing it for cultural or aesthetic import. In John Casti's popular book *Complexification: Explaining a Paradoxical World through the Science of Surprise*, he explains how the science of complexity attempts to reduce, capture, and manage surprise in nature, creating models that render the unexpected predictable. As Casti writes, the "science of surprise" attempts "to understand *why* surprises occur and whether or not there's anything we can do about them." [32] Or as John von Neumann put it, "All stable processes we shall predict. All unstable processes we shall control." [33 We might argue similarly for realms of aesthetic experience: we may expect surprise in general, but still be interested in, and, perhaps, moved by, the particular details of that surprise when it manifests. Indeed, generative art might manage surprise like the science of surprise, allowing and encouraging surprise to occur in a controlled, delimited sphere that is unthreatening, just as capitalism manages novelty and newness by allowing it to appear in such a way that it does not threaten the system that produces it. Generative art also encourages both author and audience to expect surprise—indeed, trains them to anticipate a coming surprise—but by doing so defuses the potentially radical nature of that surprise.

Perhaps we have been trained to be surprised even when, or especially when, we are expecting surprise. One only needs to think of current film thrillers to illustrate this point. We know that a surprise or twist awaits us at the end. We have been trained to read these surprises from the very beginning of a narrative, and much of the pleasure that arises in contemporary film and television comes from trying to figure out exactly what form such surprise will take. We also know that we are being held in suspense and that our surprise is being managed. Alfred Hitchcock famously privileged suspense over surprise, where suspense renders the coming surprise useful. But if we are expecting the unexpected, how then can we be 'taken' by the surprise that appears at the end? [34] The film scholar Tom Gunning has argued that "Modernity must partly be understood as learning to be surprised by certain innovations, a discourse which valorizes and directs our attention to such changes and the excitement they can provoke." [35] Capitalist modernity might be defined as a systemic process that trains us to be excited by changes in what is already familiar. How can we still be surprised by the constant cycle of creative destruction that is capitalism? How does the recent iPhone slogan, "The biggest thing to happen to iPhone since iPhone," entice us? [36] This is the modality of 'surprise, surprise'—the repetition of change that is expected and predictable becomes pleasurable even as the possibility of the unexpected is displaced. How much information (in the theoretical sense) is produced in the movement from one iPhone to the next? Capitalist modernity trains us to be surprised by the repackaging of the old as something new, to expect surprise yet nevertheless be surprised in spite of our expectations.

Gunning emphasized that events such as World Expositions rejuvenated surprise in a world where, borrowing from Georg Simmel, the experience of individuals in modern society had become blasé. For example, Gunning describes how World Expos "cued visitors to experience astonishment. The discourse of modernity, then, is not only one of innovation, but precisely one of novelty, maximizing the dazzling experience of the new." [37] Such an observation is not foreign to practitioners of contemporary digital art. Richard Wright argued in his article "New Media, Old Technology" that art and technology exhibitions during the nineteen eighties and nineties actively sought artworks using "cutting edge technology," regardless of their aesthetic import. [38] Such exhibitions were meant to display the latest technology instead of portraying the fruits of a critical art practice. This emphasis, for Wright,

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> implies that to work in the newest media you need the newest technology. The effect is to divert attention from innovations in currently used media by implying that artists can only retain their radical credentials by concentrating on the 'cutting edge' of new technology. And, *surprise, surprise*, it is exactly this mythic trajectory of technology that commercial companies depend on to motivate the consumption of their endless releases of new products that allow you do the same thing more often. [39]
> **[BLOCK QUOTE ENDS]**

While such observations are in line with Gunning concerning the excitement for new technology under capitalism (here, transported to the context of technology and art exhibitions), we are concerned primarily with the formal structures of generative art, not the cultural contexts in which these artworks are presented. Generative art, we argue, has gone further and internalized this process—the 'make it new' of aesthetic modernism and capitalist modernity—within the artwork itself. This internalization of novelty can be appreciated in the arguments of some generative artists that it is not enough for an artistic algorithm to be generative of novel outputs, but that the process itself must be novel. For example, in 2012, Marius Watz tweeted: "Temporarily banned algorithms: Circle packing, subdivisions, L-systems, Voronoi, the list goes on. Unless you make it ROCK, stay away." [40] The tired re-use of certain generative algorithms neutralizes their surprising elements and can only be justified through the spectacular—thus, making old algorithms "ROCK" plugs directly into electric discourses of astonishment to refresh and rejuvenate the old.

Heading: Beyond Surprise for Surprise's Sake

While some embrace postmodernity as an abandonment of the grand narratives of modernity such as the rise of industrial capitalism and alienation, others view it as yet another intensification of such processes. In this latter scenario, the training of individuals to be astonished, to be surprised, to constantly expect the unexpected, escalates to unprecedented levels. Taken in this light, the focus on surprise in generative art can be seen to reflect a culture driven by the need for constant renewal. While generative artworks may not explicitly train us to be surprised, they reflect back to us an aesthetic experience, for both author and audience, that mirrors the pleasures of a culture driven by the creative destruction of neoliberal capital. In this scenario, the search for surprise in generative art cannot be explained only by a drive toward the 'natural,' or as a consequence of complexity science, but also through the influence of dominant socio-economic paradigms.

The argument that surprise is intimately related to contemporary modes of production provides a means for understanding the algorithms employed by generative art, but it has the added benefit of reminding us that generative art is influenced by social and economic factors. Thus we can imagine varieties of critically-aware generative practice that investigate these connections more concretely. Rather than uncritically seeking the generation of surprising effects—for example using models of "naturally occurring processes beyond the influence of culture and man," as Galanter put it—alternative approaches may be productive to explore. [41] One such alternative explores algorithms as 'part of', rather than 'apart from', contemporary life. Algorithms are created by humans and it is clear that their role in contemporary society will continue to grow in both magnitude and complexity. [42] As David Berlinski tells us, "it has been the algorithm that has made possible the modern world." [43] Similarly, Kevin Slavin describes how contemporary algorithms "acquire the

sensibility of truth by the simple force of repetition and human exposure. They become naturalized, they ossify and calcify. They embed themselves into reality, and shape it." [44]

But while we may objectively recognize the constitutive role algorithms have come to play in the contemporary world, we subjectively experience it most acutely in the surprising moments when they break down. For example, while Twitter's trending algorithm is considered an important factor in the circulation of public knowledge, it is only questioned critically when it deviates from how we expect it to function; e.g., when #OccupyWallStreet trended in other cities around the world, but not in New York City where the movement was centered. [45] As Tarleton Gillespie writes: "…in many ways, algorithms remain outside our grasp, and they are designed to be. This is not to say that we should not aspire to illuminate their workings and impact. We should. But we may also need to prepare ourselves for more and more encounters with the unexpected and ineffable associations they will sometimes draw for us…." [46]. The algorithms that Gillespie references are complex and unpredictable, not because they model complex natural systems, but because they operate in complex social environments. They produce surprise when their outputs fail to correspond to our 'natural' expectations about how they should function. [47] Trending algorithms, search services, artificial intelligence classifiers, etc.—we take all of these for granted; that is, until they surprise us with an unexpected result, causing us to reflect on their ideological biases.

So how then might a more critically-reflective generative art practice look; one that locates its aesthetics in such prescient moments of algorithmic breakdown? Examples of such socially and politically engaged work can already be found: Heather Dewey Hagborg's *Stranger Visions* (2012), where sculptural 3D models of faces are computationally generated from publicly collected DNA; Julian Oliver and Danja Vasiliev's *Men in Grey* (2009-2014), where private text, images, and sounds are extracted from public WiFi systems and displayed to the public on the group's altered briefcases; !Mediengruppe Bitnik's *Random Darknet Shopper* (2014-present), where a bot algorithmically selects and purchases products available on the Darknet, which are then shipped directly to an art gallery for exhibition; Zach Blas and Jemima Wyman's *im here to learn so :))))))* (2017), an installation that interrogates machine learning algorithms by appropriating Microsoft's AI chatbot Tay, which spewed racist and misogynist material that it had learned; even the New York Times Research & Development group's *Openpaths* (2011), which provides a database and visualization tool for the public to examine their 'private' location data, after the discovery that Apple devices were surreptitiously collecting this data. In each of these projects, computational systems embrace surprise to investigate forms of algorithmic control and management that operate without our knowledge or examination. They help us move past the 'surprise, surprise' of a generative art that simply reproduces the faux novelty of neoliberal capitalist production and toward an aesthetics of surprise that illuminates the often opaque, algorithmic influences permeating contemporary society.

REFERENCES AND NOTES

[1] Philip Galanter, "What is Generative Art? Complexity Theory as a Context for Art Theory," in *Proceedings from GA2003–6th Generative Art Conference* (2003), 9. For other analyses of Galanter's claim about ideology see Geoff Cox. *Antithesis: The Dialectics of Software Art* (Århus, Denmark: Digital Aesthetics Research Center, University of Aarhus, 2010), 23. And, Karen Cham and Jeffrey Johnson, "Complexity Theory: A Science of Cultural Systems?" *M/C Journal* 10, No. 3 (June 2007), http://journal.media-culture.org.au/0706/08-cham-johnson.php (accessed August 15, 2019).

[2] Philip Galanter, "What is Generative Art? Complexity Theory as a Context for Art Theory," 19. It should be noted that ideology also attempts to 'naturalize,' rendering culture and its aesthetic productions as 'natural,' unchanging, and beyond the reach of 'man.' In terms of generative art, the use of the science of complexity may well serve a similar function, naturalizing the subjective choices of artists, and placing them beyond the reach of cultural critique.

[3] Adam Smith. *Essays on Philosophical Subjects,* ed. W.P.D. Wightman, J.C. Bryce, and I.S. Ross (Oxford: The Oxford University Press, 1980), 33.

[4] Ibid., 35.

[5] Matt Pearson, *Generative Art: A Practical Guide Using Processing* (Shelter Island, NY: Manning Publications Co., 2011), 6. One could continue to expand the references to surprise as essential to the generative method and privileged within the discourses surrounding generative art. For example, see Jon McCormack and Alan Dorin, "Art, Emergence, and the Computational Sublime" in proceedings from Second Iteration: A Conference on Generative Systems in the Electronic Arts (Melbourne, Australia: CEMA), 67-81. Charles Hartmann, *The Virtual Muse: Experiments in Computer Poetry.* (Hanover, NH: University Press of New England, 1996), 35-56. Or, see the explanation of the motivations behind Eva Schindling and Daniel Wilson's interactive installation "Deterministic Nonperiodic Flow (1963)" in Eva Schindling and Daniel Wilson, "Deterministic Nonperiodic Flow (1963)," http://www.evsc.net/media/dnf.pdf (accessed August 15, 2019), 2.

[6] Rafael Lozano-Hemmer, "Metaphors of Participation," Interview by Heimo Ranzenbacher in *Ars Electronica 2001: Takeover - Who's Doing the Art of Tomorrow*, eds. Stocker, G. and C. Schöpf (Springer, Wein, 2001), 243.

[7] Inke Arns, "Read_me, Run_me, Execute_me: Software and Its Discontents, or: It's the Performativity of Code, Stupid!" in *read_me: Software Art & Culture*, eds. Olga Goriunova and Alexi Shulgin (Århus, Denmark: University of Aarhus, 2004), 178. Of course, ceding some level of authorial control is an important aspect of generative art. We have chosen to focus more closely on the idea of surprise and must leave aside the question of authorial intention. This topic has been thoroughly examined elsewhere, for example, by Geoff Cox and Adrian

Ward, "How I Drew One of My Pictures, or the Authorship of Generative Art" in the *Proceedings from GA1999, Generative Art Conference*, Politecnico di Milano, Italy (Rome, Italy: Editrice Librerie Dedalo, 1999).

[8] Philip Galanter, "What is Generative Art? Complexity Theory as a Context for Art Theory,", 4.

[9] Ibid., 7.

[10] Christopher G. Langton, "Studying Artificial Life with Cellular Automata," *Physica D*, 22, (1986), 120-149.

[11] Stephen Wolfram, "Universality and Complexity in Cellular Automata," *Physica D*, 10, (1984), 1-35.

[12] Christopher G. Langton, "Studying Artificial Life with Cellular Automata," 130.

[13] In "Computing Machinery and Intelligence," Alan Turing discussed the possibility of algorithmic surprise, remarking that "Machines take me by surprise with great frequency." This happens, he says, when "I do not do sufficient calculation to decide what to expect them to do, or rather because, although I do a calculation, I do it in a hurried, slipshod fashion, taking risks." Thus, for Turing, it is not that algorithms are capable of producing something new, but rather that the unexpected arises because human thought is fallible –– the human mind fails to comprehend the consequences of an algorithm or process. Alan Turing, "Computing Machinery and Intelligence," in *The New Media Reader*, eds. Noah Wardrip-Fruin and Nick Montfort (Cambridge, MA: The MIT Press, 2003), 50-64. Vera Molnar references similar experiences in a more positive light when she says: "So you have to prepare yourself to be surprised, which one wouldn't have done working by hand or otherwise. And on the other hand mistakes are also good surprises. You get a comma or slash wrong and something comes that you hadn't wished for at all." https://vimeo.com/273642211#t=905s

[14] Marius Watz, "Fragments on Generative Art," *Vague Terrain* 03 (June 2006), https://www.artengine.ca/electricfields/2010/vagueterrain-watz-en.php (accessed August 15, 2019). Watz writes, "The artist specifies the initial boundaries and strategies of creation, and then enters into a feedback loop of adjusting parameters in a search for optimal regions in parameter space. The moment of genuine surprise is often the moment of breakthrough." Elsewhere we have unpacked the terminology of 'genuine' in relation to the notion of generative art since both terms carry etymological connections to words for begetting or engendering. Indeed, we also analyzed the differences and similarities between other words within this etymological matrix such as 'engine' and 'genius'—both which focus on the process of engendering change although the former is associated with the machine and the latter with the human. For more on the relationship between the engine and the genius, see Daniel Howe and Braxton Soderman, "The Aesthetics of Generative Literature: Lessons

from an Electronic Writing Workshop," *Hyperrhiz: New Media Cultures*, 6 (October 2009), http://hyperrhiz.io/hyperrhiz06/essays/the-aesthetics-of-generative-literature-lessons-from-a-digital-writing-workshop.html (accessed August 15, 2019).

[15] Philip Galanter, "What is Generative Art? Complexity Theory as a Context for Art Theory," 8.

[16] Jeanne Willette, "Modernist Painting" by Clement Greenberg," https://arthistoryunstuffed.com/modernist-painting-by-clement-greenberg (accessed August 15, 2019).

[17] José Van Dijck, "Datafication, dataism and dataveillance: Big Data between scientific paradigm and ideology." *Surveillance & Society* 12.2 (2014): 197-208.

[18] Jon McCormack et. al. "Ten Questions Concerning Generative Computer Art," *Leonardo*, Vol. 47, No. 2, (2014), 135–141.

[19] Philip Galanter, "What is Generative Art? Complexity Theory as a Context for Art Theory," 9. Matt Pearson, *Generative Art: A Practical Guide Using Processing*, 9. Pearson's statement is a slight modification of Galanter's slogan that "generative art is as old as art itself." See Philip Galanter, "'Generative art is as old as art'. An interview with Philip Galanter," in artificial.dk (September, 2004), http://www.artificial.dk/articles/galanter.htm (accessed August 15, 2019). In another example, Jon McCormack et. al, write that "generative procedures have a long history in art that predates the computer by thousands of years." Jon McCormack et al., "Ten Questions Concerning Generative Computer Art," 135.

[20] While Watz reminds us that generative art has a deep history—"…the aesthetic application of rules comes with a wealth of historical precedents, dating back as far as humans have been known to employ scientific principles"—he also does a fine job of historicizing 'computational' production of generative art in terms of cultural, economic, social and technological factors. Marius Watz, "Closed Systems: Generative Art and Software Abstraction," http://mariuswatz.com/2012/03/30/closed-systems-generative-art-and-software-abstraction/ (accessed May 15, 2016).

[21] One might mention a few key texts contributing to this process of historicizing the digital. See Jay David Bolter and Richard Grusin, *Remediation: Understanding New Media*. (Cambridge, MA: MIT Press, 2000). See Lev Manovich, *The Language of New Media*. (Cambridge, MA: MIT Press, 2001).

[22] Philip Galanter, "'Generative art is as old as art'. An interview with Philip Galanter."

[23] For example, one might consider Espen Aarseth's use of this strategy to debunk the hyperbolic claims of hypertext's supposed newness with the more analytic and historically expansive concept of cybertext—a concept that would even draw the ancient I-Ching into

its purview. See Espen Aarseth, *Cybertext: Perspectives on Ergodic Literature* (Baltimore, MD: Johns Hopkins University Press, 1997).

[24] Philip Fisher places many aesthetic categories firmly within what he calls the aesthetics of surprise: "both wonder and the sublime are also categories within the aesthetics of surprise and the sudden, as is that favorite modern aesthetic category, shock." Philip Fisher, *Wonder, The Rainbow, and the Aesthetics of Rare Experiences* (London: Harvard University Press, 1999), 1.

[25] Longinus, *On Great Writing (On The Sublime)*, trans. by G.M.A. Grube (Indianapolis, IN: Hackett Publishing Company, 1991), 4.

[26] Philip Fisher, *Wonder, The Rainbow, and the Aesthetics of Rare Experiences*, 1.

[27] Jon McCormack and Alan Dorin, "Art, Emergence, and the Computational Sublime."

[28] Ibid., 12.

[29] McCormack et al., "Ten Questions Concerning Generative Computer Art," 135.

[30] John L. Casti, *Complexificiation: Explaining a Paradoxical World Through the Science of Surprise* (New York, NY: HarperPerennial, 1994), 3.

[31] Ludwig Wittgenstein, *Culture and Value*, trans. Peter Winch (Chicago: University of Chicago Press, 1984), 45.

[32] John L. Casti, *Complexificiation: Explaining a Paradoxical World Through the Science of Surprise*, 3.

[33] Quoted in Freeman Dyson, *Infinite In All Directions* (New York, NY: Harper & Row, 1988), 182.

[34] Alfred Hitchcock famously differentiated between suspense and surprise, arguing that suspense—allowing the audience to know of an impeding event in the film in order that they may "[participate] in the scene"—should be the form chosen over surprise, "Except when the surprise is a twist, that is, when the unexpected ending is, in itself, the highlight of the story." François Truffaut and Helen G. Scott, *Hitchcock* (New York, NY: Simon and Schuster Paperbacks, 1983), 73.

[35] Tom Gunning, "Re-newing Old Technologies: Astonishment, Second Nature and the Uncanny in Technology from the Previous Turn-of-the-Century," lecture presented at MIT (February, 1998), http://web.mit.edu/m-i-t/articles/index_gunning.html (accessed August 15, 2019).

[36] Apple's official Web Site, "iPhone," http://www.apple.com/iphone/ (accessed February 20, 2013).

[37] Tom Gunning, "Re-newing Old Technologies: Astonishment, Second Nature and the Uncanny in Technology from the Previous Turn-of-the-Century."

[38] Richard Wright, "New Media, Old Technology," *Variant* 2, No. 6 (Autumn 1998), http://www.futurenatural.net/ (accessed August 15, 2019).

[39] Ibid.

[40] Quoted in Bruce Sterling, "Tech Art: Algorithm Criticism," *Wired* (February 16, 2012), http://www.wired.com/2012/02/tech-art-algorithm-criticism/ (accessed August 15, 2019).

[41] It is worth invoking Frieder Nake's article "There Should Be No Computer Art" from 1971 where he explicitly lamented the attachment of computer-generated art to the false novelty of capitalist commodification. Instead of being "[interested] in producing some more nice and beautiful objects for computers" he argued that "the interest in computers and art should be the investigation of aesthetic information as part of the investigation of communication. This investigation should be directed by the needs of the people." Frieder Nake, "There Should Be No Computer Art," *Bulletin of the Computer Arts Society* (1971), 19.

[42] Here we are thinking of Tarelton Gillespie's critical analysis of what he calls "public relevance algorithms"—algorithms that fundamentally condition how we understand and gain knowledge about the world today (e.g. search, recommendation, trending algorithms, etc.). Tarleton Gillespie, "The Relevance of Algorithms" in *Media Technologies: Essays on Communication, Materiality, and Society* (Cambridge, MA: The MIT Press, 2014), 167-194.

[43] David Berlinski, *The Advent of the Algorithm: The 300-Year Journey from an Idea to the Computer* (San Diego, CA: Harcourt, 2001), XV.

[44] Kevin Slavin, "How Algorithms Shape Our World," (Presentation, *Ted Talk*, July 2011), http://www.ted.com/talks/kevin_slavin_how_algorithms_shape_our_world?language=en (accessed August 15, 2019).

[45] For an analysis of the Twitter Trending algorithm that appeared after unexpected, and surprising, omissions during the Occupy Movement, see Gilad Lotan, "Data Reveals That "Occupying" Twitter Trending Topics is Harder Than it Looks!," *Gilad Lotan: musings on data, visualization and politics* (blog), October 12, 2011, http://giladlotan.com/2011/10/data-reveals-that-occupying-twitter-trending-topics-is-harder-than-it-looks/ Also see, Tarleton Gillespie, "Can an algorithm be wrong? Twitter Trends, the specter of censorship, and our faith in the algorithms around us," *Culture Digitally* (blog), October 19, 2011, http://culturedigitally.org/2011/10/can-an-algorithm-be-wrong/ (accessed August 15, 2019).

[46] Tarleton Gillespie, "The Relevance of Algorithms," 192.

[47] Gillespie provides another example of unexpected realization of an algorithm's functioning because of the complexity of Amazon's informatic systems: "In 2009, more than fifty-seven thousand gay-friendly books disappeared in an instant from Amazon's sales lists, because they had been accidentally categorized as 'adult.' Naturally, complex information systems are prone to error. But this particular error also revealed that Amazon's algorithm calculating 'sales rank' is instructed to ignore books designated as adult." Ibid., 171.

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